



ACADEMIC CITY
UNIVERSITY

BSc

ELECTRICAL & ELECTRONICS ENGINEERING

Electrical and Electronics Engineering at ACity

Electrical and Electronics Engineering students at ACity engage in a technical discipline featuring the in-depth study, design, and application of equipment, devices, and systems that use electricity, electronics, and electromagnetism for the development of groundbreaking products. Our focus is to produce versatile graduates who seamlessly blend theoretical and practical know-how to solve contemporary and involving problems in electrical and electronics engineering.



Be Ready for an Electrifying Future!

Did you know “the first practical application of electricity was the telegraph, invented by Samuel F.B. Morse in 1837?”

Did you know “the need for electrical engineers was not felt until the invention of the telephone (1876) by Alexander Graham Bell and of the incandescent lamp (1878) by Thomas A. Edison?”

Since the post-world war years, Electrical and Electronics Engineering programs have made a significant foray into an interdisciplinary field of studies combining electronics, telecommunication, computer science and engineering, robotics, aerospace engineering, biomedical engineering, and artificial intelligence. Currently, active research areas include applied electromagnetics, biomedical engineering, power and energy systems, signal and image processing, wireless communications, and unmanned systems. Definitely, the field is set to direct the future of industry.

Top Careers in Electrical And Electronics Engineering

There is a wide range of career opportunities available for students of Electrical and Electronics Engineering. Be it in telecommunications or energy, there is space for everyone. Here are some top careers in the field:

- ✦ Computer Hardware Engineer
- ✦ Biomedical Engineer
- ✦ Aerospace Engineer
- ✦ Electrical Engineer
- ✦ Systems Analyst

Why ACity?

Home to
students from over
23 countries
across Africa

A fully digitized
campus with the fastest
internet speed
in a Ghanaian university

70%
practical lessons

**Experienced
Faculty**
with Industry Expertise

Every **ACitizen** has
the opportunity to do a
work placement
every vacation

Student-To-Faculty Ratio
5:1

Houses a
Digital Makerspace
and a **Technology &
Entrepreneurial Center**
for **Creative freedom
& Innovation**

Our Unique Learning Pillars



Experiential Learning

Hands-on learning to prepare students to readily apply concepts, to easily integrate into the workspace.



Contextual Learning

Solving real grass-root problems to expose students to the local context and develop empathy towards the continent's progress.



Unified Learning

A project-based approach that combines concepts across courses to connect the dots and enable unified learning.



Extensional Learning

Arms students with a viable toolkit to help them confront real-life issues, they may not have encountered during their academic life, squarely.

Entry Requirements

WASSCE/NECO

Minimum C6 in 6 subjects including 3 core subjects (English, Mathematics, Integrated/General Science) and 3 elective subjects. (Physics, Elective Mathematics + Chemistry OR any other elective relevant to the chosen Programme)

SSCE

Minimum D or a pass in 6 subjects including 3 core subjects (English, Mathematics, Integrated/General Science) and 3 elective subjects. (Physics, Elective Mathematics + Chemistry OR any other elective relevant to the chosen Programme)

IGCSE O-LEVEL & A-LEVEL

Minimum of 5 credit passes in the IGCSE/O-Levels (English and Maths mandatory) and 3 passes in the A-Levels. (Elective/Add/Further Maths and Physics mandatory)

ENGLISH IB

Minimum of 5 credit passes in the IGCSE/O-Levels (English and Maths mandatory) and a minimum score of 4 points in 3 Higher Level (HL) subjects. (Elective/Add/Further Maths and Physics mandatory)

FRENCH IB

Minimum of 50% overall average pass. (subject to approval NAB) (Maths, English and Physics mandatory)

AMERICAN HIGH SCHOOL

Minimum GPA of 3.0 (Maths, English and Physics mandatory)

COURSE OUTLINE

SEMESTER 1

Course Name

Communication Skills
French Language
Fundamentals of Innovation and Entrepreneurship (FIE) Seminar I
Introduction to Engineering
Introduction to Programming with Python
Physical Sciences
Pre-Calculus (with MATLAB)
Technology and Society

SEMESTER 2

Course Name

Analytic Geometry and Calculus I (with MATLAB)
Introduction to Electrical and Electronics Engineering
Fundamentals of Innovation and Entrepreneurship (FIE) Seminar II
Introduction to Multidisciplinary Design
Logic and Critical Thinking
Programming in C
Sensors, Measurements and Instrumentation
Text and Meaning

SEMESTER 3

Course Name

Analytic Geometry and Calculus II (with MATLAB)
Circuit Theory
Electrical Energy Conversion and Green Electronics
Electromagnetic Fields and Waves Theory
Fundamentals of Innovation and Entrepreneurship (FIE) I
Introduction to Material Science and Engineering
Leadership Seminar I

SEMESTER 4

Course Name

African Studies
Applied Linear Algebra (with MATLAB)
Electrical Machines
Fundamentals of Innovation and Entrepreneurship (FIE) II
Introduction to Contemporary Electric Power Systems
Probability, Statistics and Reliability (with MATLAB)

SEMESTER 5

Course Name

Differential Equations (with MATLAB)
Digital Systems Design
Leadership Seminar II
Microelectronics
Numerical Methods (with MATLAB)

SEMESTER 6

Course Name

Automatic Control Systems
Electric Power Generation, Transmission and Distribution
Embedded Systems
Industry Internship
Mechatronics

SEMESTER 5

Course Name

Power Electronics
Signals and Systems
Systems Dynamics

SEMESTER 6

Course Name

Principles of Communication Systems
Project Management, Engineering Economics and Risk Analysis

SEMESTER 7

Course Name

Engineering Thermodynamics
Probabilistic Systems Analysis and Applied Probability
Project Phase I
Technical Elective - I
Technical Elective - II

SEMESTER 8

Course Name

Environmental Science and Engineering
Professional Ethics and Values
Project Phase II
Technical Elective - III
Technical Elective - IV

ELECTIVES

Renewable Energy Systems

Smart Grid
Solid State Electronic Devices
Energy Management for a Sustainable Future
Green Electronics

Radio Frequency (RF) and Microwave Circuits Design

Microwave Engineering
Radio Frequency (RF) Circuits Design
Optoelectronic Devices
VLSI Design

Intelligent Systems: Robotics and Mechatronics

Advanced Control Systems
Industrial Automation and Robotics
Artificial Intelligence and Robotics
Modeling and Analysis of Mechatronic Systems

Intelligent Systems: Controls and Automation

Advanced Control Systems
Computer Architecture and Organisation
Artificial Intelligence and Robotics
Modeling and Analysis of Mechatronic Systems

Power Systems Engineering

Advanced Power Generation, Transmission and Distribution
Power System Operation and Control
Electric Power Systems Analysis and Design
Power System Protection and Switchgear

Biomedical Engineering

Biomedical Instrumentation
Biomedical Signal Analysis
Real-Time Digital Signal Processing
Wireless Networks